

1. Autocorrelation Function (ACF)

- ACF measures the linear relationship between observations of a time series separated by a lag of k .
- Formula:

$$\rho(k) = \frac{Cov(Y_t, Y_{t-k})}{Var(Y_t)} = \frac{E((Y_t - \mu)(Y_{t-k} - \mu))}{\sigma^2}$$

Where $\rho(k)$ is the autocorrelation at lag k , μ is the mean of the time series and σ^2 is the variance of the time series.

2. Partial autocorrelation Function (PACF)

- PACF measures the direct correlation between Y_t and Y_{t-k} , removing the influence of intermediate lags (from lag 1 to lag $k - 1$).
- Formula:

$$\rho_{kk} = Corr(Y_t, Y_{t-k} | Y_{t-1}, Y_{t-2}, \dots, Y_{t-k+1})$$

3. MA model

An MA model uses past forecast errors in a regression model. In the MA(q) model, the current value of the series depends on the previous q errors terms. The equation is:

$$Y_t = e_t + \beta_1 e_{t-1} + \beta_2 e_{t-2} + \dots + \beta_q e_{t-q}$$

Where $\beta_1, \beta_2, \dots, \beta_q$ are parameters, e_t is white noise.

- **The code of the MA(1) and MA(2) model in Stata**

//MA(1)

```
gen y2 = 0
forvalues i = 1(1)500 {
    gen e = rnormal(0,1)
    replace y2 = e if t == 1
    replace y2 = 0.8*L1.e + e if t > 1
    mkmata t y2 e, matrix(serie)
}
label var y2 "MA(1)"
drop e
ac y2
pac y2
```

//MA(2)

```
gen y3 = 0
forvalues i = 1(1)500 {
    gen e = rnormal(0,1)
    replace y3 = e if t == 1
    replace y3 = 0.8*L1.e - 0.02*L2.e + e if t > 1
    mkmata t y3 e, matrix(serie)
}
```

```

}
label var y3 "MA(2) "
drop e
ac y3
pac y3

```

4. AR model

The AR model describes a time series in terms of its own previous values. In an AR(p) model, the current value of the series is based on a linear combination of the previous p values plus a random error term. The equation is

$$Y_t = \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \dots + \alpha_p Y_{t-p} + e_t$$

Where $\alpha_1, \alpha_2, \dots, \alpha_p$ are parameters and e_t is white noise.

- **The code of the AR(1) with $\alpha_1 = 2$ and AR(2) with $\alpha_1 = 0.2$ and $\alpha_2 = 0.3$ model in Stata**

```

//AR(1)
gen y7 = 0
forvalues i = 1(1)500 {
    gen e = rnormal(0,1)
    replace y7 = e if t == 1
    replace y7 = 2*L1.y7 + e if t > 1
    mkmat t y7 e, matrix(serie)
}
label var y7 "AR(1) with phi = 2"
drop e
ac y7
pac y7

//AR(2)
set obs 50
gen t = _n
tsset t
gen y8 = 0
forvalues i = 1(1)50 {
    gen e = rnormal(0,1)
    replace y8 = e if t == 1
    replace y8 = 0.2*L1.y8 + e if t == 2
    replace y8 = 0.2*L1.y8 + 0.3*L2.y8 + e if t > 2
    mkmat t y8 e, matrix(serie)
}
label var y8 "AR(2) "

```

```
drop e
ac y8
pac y8
```

5. ARMA model

An ARMA(p,q) model combines both AR and MA components, modeling the series using both past values and past errors. The model can be written as

$$Y_t = \alpha_1 Y_{t-1} + \alpha_2 Y_{t-2} + \dots + \alpha_p Y_{t-p} + e_t + \beta_1 e_{t-1} + \beta_2 e_{t-2} + \dots + \beta_q e_{t-q}$$

- **The code of the ARMA(2,2) model in Stata is**

```
//ARMA(2,2)
set obs 500
gen t5=_n
tsset t5
gen y5=0
forvalues i = 3(1)500 {
    replace y5=0 if t5==1
    gen e = rnormal(0,1)
    replace y5 = 0.7*L1.y5 + e +0.6*L1.e if t5 == 2
    replace y5 = 0.7*L1.y5 +0.2*L2.y5 + e + 0.6*L1.e -
    0.08*L2.e
    mkmat t5 y5 e, matrix(serie)
    drop e
}
ac y5
pac y5
```